Experiment - 6

Error detection and Error correction

Aim :

Write Program to implement error detection and detection using hamming code concept Make test run to input data stream and verify error correction feature

Sender Program :

* Input to sender file should text any length Program should convert text to binary
* Apply hamming code concept on the binary data and add redundant bits to it
* Save in file called channel

code (sender.py)

```
data = list(map(int,input("Enter data bits : ").strip());
m = len data()
r=0
while (2**r) < (m+r+1)
    r+=1
n=m+r
hamming = [0]*(n+1)
j=0
for(i in range(1,n+1);
    if(1*(i-1)) != 0;
```

```
    hamming[i] = data[j]
    j+=1
for p in range(r):
    pos = 2**p
    parity = 0
for i in range(1, r+1):
    parity = hamming[i]
    hamming[pos] = parity
    code = hamming[1]
    print("Data bits:", data)
    print("hamming code:", code)
```

output:

```
Enter data bits: 1011
Data bits: [1,0,1,1]
Hamming code: [0,1,1,0,0,1,1]
```

Recive code

```
received = list(map(int, input("
        Enter received hamming
code:").strip()))

n = len(received)
r = 0
while (2**r) < n:
    r+=1
error pos = 0
for p in range(r):
```

```
        pos = 2**p
        parity = 0
    for i in range (1,n+1)
     if i in range (1,n+1)
    data.append (received (i-1))
    Print ("original data list: data);
```

Output:

Suppose 0110011 change to 0111011

error detected at position : 4

corrected hamming code : [0,1,1,0,0,1,1]

original data bits : [1,0,1,1]

Result :

Thus, the hamming code algorithm was successfully implemented in Python, the sender generated the hamming code from the given data, the reciever successfully.